AI REPORT P2



INTRODUCTION

The objective of this lab assignment

Functions

minimax

```
minimax(self, currentState, depth, whitesTurn)
```

gameMinimax()
Canviar nom a auxMinimax()

Questions

MINIMAX

1. Whites start moving. Implement the dynamics of a game in which both, whites and blacks, follow the same Minimax algorithm to try to check-mate each other. Assume that both implement minimax with a depth of 4 moves (2p)

a. Once implemented, run the same game a few times. How often do whites win?

With a depth of 4 moves for both players the game frequently reaches the maximum number of moves allowed without a checkmate (we fixed it in 100), therefore, ends in a tie.

1: ties
2: ties
3: ties
…
Limit of plays: ties
The number of white victories is low ≃ zero.

White pieces move the rook and the king. Black piexes move the king and after,they start repeating movements. Finally we arrive to maxMoves = 100 and the program declares ties.

This means that the two algorithms have gotten into a loop of movements.

b. Provide a justification for that.

The minimax algorithm is determinant: if it starts from the same state and uses the same heuristic and depth, it will always make the same decision (while the order in which the moves are generated is also the same).
Both players use the same algorithm. Moreover, the initial position is always the same, so they will probably choose the same movements. In addition, the depth is limited (4 for both players in this case). And they only consider four individual movements in each search.

We have set the limit of our game at 100 movements. If the checkmate is not found before the limit of movements, the program declares 'false' ties. The heuristics also influences: the evaluation doesn't reflect perfectly the positions that lead to checkmate and the players repeat movements. We would have liked to improve the heuristics but we did not have enough time.

Besides that, the heuristic favors White when they capture the black rook.
When the black rook is removed, the heuristic of the basecode adds 50 points and then evaluates the distance between the kings, as well as the position of the black king in relation to the edges of the board.

On the other hand, when the white rook is removed, it subtracts 50 points and penalizes the position of the white king. Therefore, the function is designed to favor a win for White and to penalize the loss of their rook.

This explains that, when we play with depth 2,2 or 1,2 White wins anyway.

2. Now run the same simulations, but varying the depth of the minimax algorithm from 3 to 4 moves both for whites and blacks. Run each possible combination of depths a few times (1.5p)

Combinations:

W:3, B:3:
Insufficient material (2kings): ties
Insufficient material (2kings): ties
Insufficient material (2kings): ties

W:3, B:4:

W:4, B:3:
Limit of plays: ties

W:4, B:4:
Limit of plays: ties

4 blancos, 1 negro: blancos. Negros 4, blancos 1:tie.

a. Plot the percentage of white wins over the total for each depth value.
Pjkrwb fjk

b. Is the result symmetric. Why?
No. White ones are favored because of the heuristics. As we have commented earlier, the heuristic evaluates positions from White's perspective, rewarding moves that benefit White and penalizing the ones that benefit Black. Therefore, even though both players use the same minimax and, in some occasions the he same depth, White may have an advantage.

Implementation Decisions

Las blancas quieren MAXIMIZAR el valor, por eso empezamos en -infinito.
Las negras quieren MINIMIZAR el valor, por eso empezamos en +infinito.

depth is the lacking moves to the depth we wanted to achieve. No els pasos que portem com a la p1.

why +depth? return [1000 + depth, None] # + depth because we want to win as fast as possible

EAT the king in boardsim so it's easier minimax. Perminir que el rey sea capturado en boardSim para que Minimax detecte que una rama ha ganado, pero impedir esa captura en board cuando se ejecuta la jugada real.

boardSim — simulación de Minimax

Permite capturar al rey contrario. Cuando desaparece de currentStateW o currentStateB, Minimax detecta que esa rama ha ganado y le asigna una puntuación muy alta.

minimax captura del rey solo en el simulador, y el tablero real se mueve en minimaxGame, justo después de decidir.

DIFerencia: turno jugador y turno arbol!

solo con 2 reyes hay ties porque no se pueden hacer jaque ni matar?

Sí, en ajedrez real es empate por **material insuficiente**: con solo dos reyes nadie puede dar jaque mate. La razón no es que "no se puedan hacer jaque", sino algo más concreto:

- **Un rey nunca puede dar jaque a otro.** Para dar jaque, una pieza debe amenazar al rey rival, pero dos reyes no pueden ocupar casillas adyacentes (sería un movimiento ilegal, porque el rey que se acercara quedaría en jaque). Así que el jaque entre reyes no existe.
- **Un rey nunca se captura.** En ajedrez real el rey no se "mata": la partida termina antes, con jaque mate. Sin otras piezas, no hay forma de forzar un jaque mate.

LEGALLITY FILTER: descart moves that leave my king on scac
deixar el rei en escac"
És fer un moviment després del qual el teu propi rei pot ser capturat pel rival. A escacs real és il·legal. Per exemple, si el teu rei està en escac per la torre rival i mous la teva torre en comptes de salvar el rei, o si poses el rei al costat de l'altre rei.
Em vaig equivocar abans. Dient que minimax ho penalitzava indirectament, vaig oblidar que tu fas continue quan un moviment captura el rei rival. Així, el rival mai captura el rei i no hi ha cap penalització: un jugador en escac pot ignorar-lo, i això distorsiona tota la partida.

CONCLUSIONS

USE OF AI